

Shor's Algorithm

*How a quantum computer finds the period of $a^x \bmod N$
— and why knowing that period factors N*

Colours carry meaning

Blue — the **counting register**, the one we measure.

Vermillion — the **target register**, where the arithmetic happens.

Purple — the **period** r , and the angles it produces.

The slides also read correctly in greyscale: colour is a reminder, never the only carrier of information.

Each act is one question and its answer

Difficulty. Opens each act: the obstacle the act has to get past.

Resolved. *Closes the act: how the obstacle was removed.*

Symbols are introduced where they are first used. A one-page table of all of them is in the backup slides.

Act I — the problem: a period that no single value reveals

Act I — the problem: find the period

Difficulty 1. Given N and a number a with $\gcd(a, N) = 1$, find the **order** r : the smallest $r > 0$ with $a^r \equiv 1 \pmod{N}$. Equivalently, r is the **period** of $f(x) = a^x \bmod N$.

Example: $N = 21, a = 2$.

x	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	...
$f(x) = 2^x \bmod 21$	1	2	4	8	16	11	1	2	4	8	16	11	1	2	4	
	$r = 6$ values, all different						then the same 6 again									

Any single value is cheap

$f(x)$ for one x costs about $n = \log_2 N$ multiplications (repeated squaring), even for a 2048-bit N .

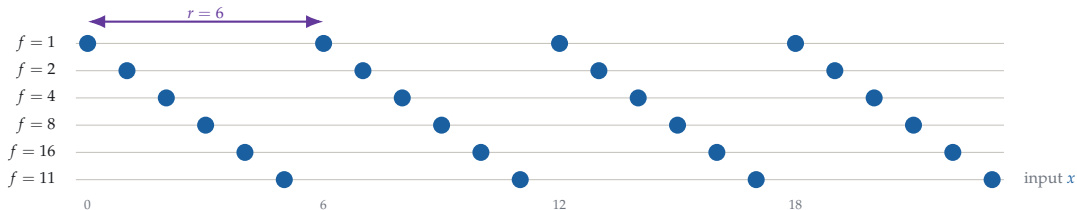
The period is not

r is a property of the *whole* sequence, not of any one value: no single $f(x)$ — nor polynomially many of them — contains it. No classical method is known that runs in polynomial time.

“A period that no single value reveals”: r is not written anywhere in the table — it is the spacing of its repetitions.

What the period does to the values: combs

The values repeat with period r and differ within one period, so $f(x) = f(x') \iff x \equiv x' \pmod{r}$. Group the inputs by the value they produce: each group is a **comb** — teeth spaced r apart, each comb with its own offset.

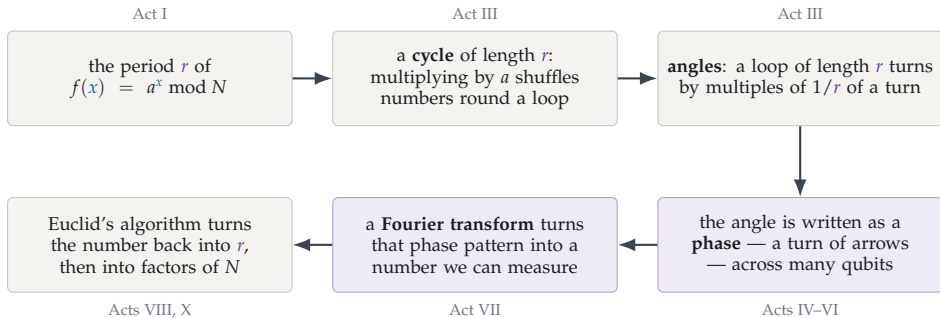


The period is geometry. Every comb has the same spacing r ; only the offset differs. Finding r means detecting a *spacing* — a job for a process that responds to the whole comb at once.

Input and output are correlated. Knowing $f(x)$ does not tell you x , but it tells you which comb x is on. Keep this correlation in mind: in Act V it becomes *entanglement*.

Why this particular period is worth the trouble — it factors N — is the finale, Act X. Until then, the period is the quarry.

The road map, in plain words



Only the two shaded steps need a quantum computer. Act II sets up the rules; Act IX explains what can silently break them. Every technical word above is defined when it first appears.

The quantum computer never computes r . It turns the period into a pattern of phases and then transforms that pattern until a single measurement reveals it.

Act II — the rules of the game

Act II — a new notation, and why

Difficulty 2. We need a model of computation in which a *global* property of many values can influence a *single* observable event. Classical probability cannot do this. Quantum mechanics can — and it is usually written in a notation built for the job.

Three reasons for a special notation — and the whole of it, on one table:

Why not just columns of numbers?

- A state of t qubits is a column of 2^t numbers. For $t = 4000$ nobody writes it out — we need a *name* for “the basis vector with label x ”: the **ket** $|x\rangle$. In $\mathbb{C}^8$: $|5\rangle = (0, 0, 0, 0, 0, 1, 0, 0)^\top$.
- The label *means* something: $|a^k \bmod N\rangle$ or $|x\rangle|f(x)\rangle$ keep the arithmetic visible; a column would hide it.
- The *shape* of an expression tells you what it is: a number, a vector, a matrix. Calculations become snapping brackets together.

The whole dictionary

symbol	name	linear algebra
$ \psi\rangle$	ket	column vector
$\langle\psi $	bra	row: conjugate transpose
$\langle\varphi \psi\rangle$	bra-ket	a number (inner product)
$ \psi\rangle\langle\varphi $	ket-bra	a matrix (outer product)
$ x\rangle y\rangle$		$ x\rangle \otimes y\rangle$, a joint label

Rows and columns are worked out on the next slides; the four rules of the model come first.

Rules 1 and 2: states and gates

(R1) A state is a unit vector of complex amplitudes

A register whose labels are $j = 0, 1, \dots, d - 1$ is in a state

$$|\psi\rangle = \sum_j \alpha_j |j\rangle, \quad \alpha_j \in \mathbb{C}, \quad \sum_j |\alpha_j|^2 = 1.$$

The α_j are **amplitudes**. One qubit: $d = 2$, $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. A register of t qubits has $d = 2^t$ labels $0, \dots, 2^t - 1$: *the dimension is exponential in the number of qubits*.

(R2) A gate is a unitary matrix

A gate maps $|\psi\rangle \mapsto U|\psi\rangle$, where U preserves length (made precise in two slides: $U^\dagger U = \mathbf{1}$), so the amplitudes still square-sum to 1. An *efficient* algorithm uses polynomially many gates, each touching one or two qubits.

Linearity is the only parallelism. $U(\sum_j \alpha_j |j\rangle) = \sum_j \alpha_j U|j\rangle$: a gate is fixed by what it does to each basis vector, and on a superposition it acts on every branch at once.

Computational basis. The vectors $|0\rangle, \dots, |d-1\rangle$ — the standard columns — labelled by integers.

Rule 3: measurement gives one label per run

(R3) The Born rule

Measuring $|\psi\rangle = \sum_j \alpha_j |j\rangle$ returns the label j with probability $|\alpha_j|^2$, and leaves the register in $|j\rangle$. **One classical label per run — never the amplitudes.** Measuring one register of a pair: the probability of a label is the squared length of the part of the vector carrying that label.

An aside: the measurement problem

Why a measurement yields one definite outcome, and what happens physically when it does, is the open question of quantum foundations known as the **measurement problem**. We do not need to settle it: the algorithm only uses R3 as a rule for the statistics.

Consequence: “compute f everywhere at once” is not enough

By linearity a quantum computer *can* prepare $\frac{1}{\sqrt{Q}} \sum_x |x\rangle |f(x)\rangle$ in one pass: f on all $Q = 2^t$ inputs. But measuring returns *one* random pair $(x, f(x))$. Repeating gives more random pairs — classical random sampling, which cannot see r (Act I).

R3 is the whole difficulty. What will save us: before the one sample is drawn, the amplitudes can be made to add up and cancel — interference — so that the sample lands where the period is.

Rule 4: two registers together

(R4) Registers combine by the tensor product

Joint basis vectors are pairs of labels, $|x\rangle |w\rangle := |x\rangle \otimes |w\rangle$; dimensions multiply. For qubits, a string of bits is read as a binary integer. In columns, $\otimes$ is the Kronecker product:

$$|1\rangle \otimes |0\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \cdot (1,0)^\top \\ 1 \cdot (1,0)^\top \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = |10\rangle = |2\rangle.$$

$\otimes$ is bilinear

$$(\lambda |a\rangle) \otimes |b\rangle = \lambda (|a\rangle \otimes |b\rangle) = |a\rangle \otimes (\lambda |b\rangle),$$

$$(|a\rangle + |a'\rangle) \otimes |b\rangle = |a\rangle |b\rangle + |a'\rangle |b\rangle.$$

A number can be moved freely between the two factors: it belongs to the *pair*, not to either register. This is the whole engine of Act IV.

Product or entangled

A **product** state $|a\rangle \otimes |b\rangle$: each register has a state of its own. Now take

$$\frac{1}{\sqrt{2}} (|0\rangle |1\rangle + |1\rangle |7\rangle) :$$

the second register's value *depends* on the first one's — like x and $f(x)$ in Act I. No choice of $|a\rangle, |b\rangle$ reproduces it. A joint state that is not a product is **entangled**: the registers are correlated, and neither has a state vector of its own.

Bra times ket: a number

A ket is a column; its **bra** is the conjugate-transpose row:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad \langle\psi| = (\bar{\alpha} \quad \bar{\beta}).$$

Row times column is the **inner product** (with $|\varphi\rangle = \gamma |0\rangle + \delta |1\rangle$):

$$\langle\varphi|\psi\rangle = (\bar{\gamma} \quad \bar{\delta}) \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \bar{\gamma}\alpha + \bar{\delta}\beta, \quad \langle\psi|\psi\rangle = |\alpha|^2 + |\beta|^2 = 1, \quad \langle j|\psi\rangle = \alpha_j.$$

Why conjugate?

Transpose alone fails over $\mathbb{C}$: for $|\psi\rangle = (1, i)^\top$ it gives $1^2 + i^2 = 0$, a nonzero vector of “length” 0. With the conjugate, $\langle\psi|\psi\rangle = \sum_j |\alpha_j|^2 \geq 0$: a genuine squared length, so Born probabilities come out real, nonnegative, summing to 1.

Reading the result

$\langle j|\psi\rangle = \alpha_j$: an amplitude is an overlap with a basis vector. The basis is **orthonormal**: $\langle j|k\rangle = 1$ if $j = k$, else 0. Two states with $\langle\varphi|\psi\rangle = 0$ are orthogonal — perfectly distinguishable by a measurement.

A bra beside a ket closes into a number: how much of $|\psi\rangle$ points along $|\varphi\rangle$.

Ket times bra: a matrix — projectors and operators

Column times row is the **outer product**, a matrix:

$$|0\rangle\langle 0| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad |1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad |0\rangle\langle 1| = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

$|0\rangle\langle 0|$ is a **projector** — not the number 0

It *keeps* the $|0\rangle$ -part of a vector and deletes the rest: $|0\rangle\langle 0|\psi\rangle = \alpha|0\rangle$. Likewise $|1\rangle\langle 1|$ keeps the $|1\rangle$ -part — it is *not* the identity. Together they rebuild everything: $|0\rangle\langle 0| + |1\rangle\langle 1| = \mathbf{1}$.

We will meet exactly this pair inside every controlled gate (Act IV).

Any matrix is a table of entries $\langle x| U |y\rangle$

$U|y\rangle$ picks column y ; $\langle x|$ then picks row x of it. So

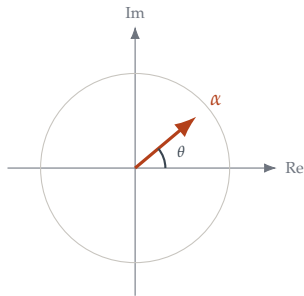
$$\langle x| U |y\rangle = U_{xy}, \quad U = \sum_{x,y} U_{xy} |x\rangle\langle y|.$$

The adjoint $U^\dagger$ (conjugate transpose) moves U across the bar: $\langle U\varphi|\psi\rangle = \langle\varphi| U^\dagger |\psi\rangle$.

Unitary, made precise. $U^\dagger U = \mathbf{1} \iff \langle U\varphi|U\psi\rangle = \langle\varphi|\psi\rangle$ for all states: U preserves every overlap and every length — total probability stays 1 (R2).

A ket beside a bra opens into a matrix. Everything ahead is rows times columns in this costume.

An amplitude is an arrow: a length and a direction



$$\alpha = |\alpha| e^{i\theta}: \text{length } |\alpha|, \text{direction } \theta$$

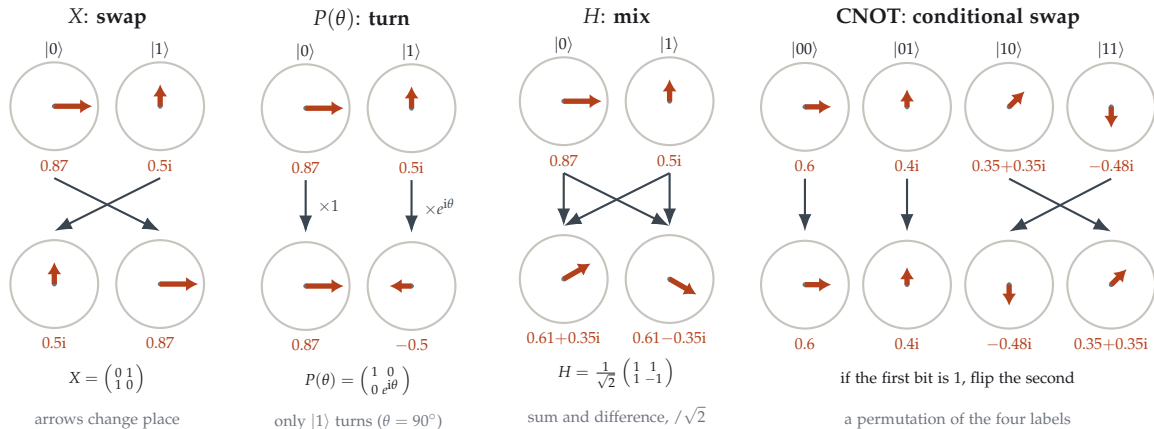
An amplitude is a complex number, so it is not a height but an **arrow** in the plane:

- its **length** $|\alpha|$ gives the probability, $P = |\alpha|^2$ (R3);
- its **direction** θ , the **phase**, is invisible to a measurement of that one label — but it decides what happens when two arrows reach the *same* label and are added (next slides).

A bar chart can only show lengths, and at best a sign. From now on each amplitude is drawn as an arrow in a clock face:  has length 0.8 and direction 40° .

Probability sees the length. Interference sees the direction. A quantum algorithm is the art of steering directions so that lengths end up where we want them.

Gates act on the whole table of arrows



A gate is one linear map on the whole table: it moves arrows (X, CNOT), turns them (P: lengths untouched), or mixes them (H) — and only mixing adds arrows, which is where interference happens.

Fact A — interference is a cross term

Suppose a final label can be reached along two paths, contributing amplitudes A_1 and A_2 . Amplitudes of paths to the same label **add, then square**:

$$P = |A_1 + A_2|^2 = (A_1 + A_2) \overline{(A_1 + A_2)}.$$

Multiply out the four products:

$$P = \underbrace{|A_1|^2 + |A_2|^2}_{\text{each path alone}} + \underbrace{A_1 \overline{A_2} + \overline{A_1} A_2}_{\text{cross term}}.$$

The two cross products are complex conjugates, so their sum is real:

$$A_1 \overline{A_2} + \overline{A_1} A_2 = 2 |A_1| |A_2| \cos \Delta,$$

with Δ the **angle between the two arrows**.

Classically, mutually exclusive alternatives add *probabilities*, $p_1 + p_2$: there is no cross term. With many paths, $|\sum_j A_j|^2$ has one cross term for every pair of paths.

The cross term is the entire difference between quantum and classical probability — and it is controlled by one thing: the relative direction of the arrows.



$\Delta = 0$: $P = 1$
constructive



$\Delta = 90^\circ$: $P = \frac{1}{2}$
no cross term: as classical



$\Delta = 180^\circ$: $P = 0$
destructive

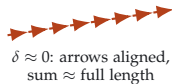
two arrows of length $\frac{1}{2}$, head to tail;
dashed: their sum

Many arrows, turning evenly: the Dirichlet kernel

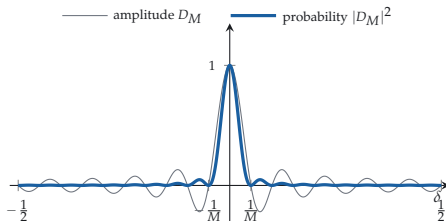
uses: Fact A

The pattern Shor's algorithm will produce: M arrows, each of length $1/M$, each turned by the same angle $2\pi\delta$ relative to the previous one. Their sum is

$$D_M(\delta) := \frac{1}{M} \sum_{x=0}^{M-1} e^{2\pi i \delta x}, \quad |D_M(\delta)| = \left| \frac{\sin \pi M \delta}{M \sin \pi \delta} \right|, \quad D_M(0) = 1.$$



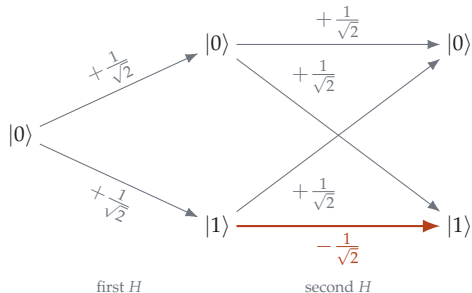
$M\delta = 1$: the walk closes



$M = 20$. Zeros wherever $M\delta$ is a nonzero integer.

One tall peak where the arrows line up, exact cancellation where the walk closes, small ripples between: this is the interference pattern of evenly turning arrows — and, squared, the entire output law of Shor's algorithm (Act VII).

Interference in the smallest example: H , then H again



A coin flipped twice

After one flip: 50/50. After two: still 50/50. Probabilities of alternatives just add.

Along a path, amplitudes **multiply**; paths ending on the same label **add** (Fact A):

$$A(|0\rangle) = \frac{1}{2} + \frac{1}{2} = 1, \quad A(|1\rangle) = \frac{1}{2} - \frac{1}{2} = 0.$$

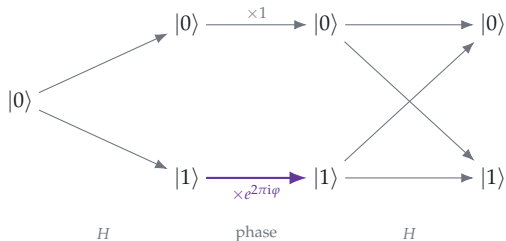
Both paths to $|0\rangle$ carry aligned arrows: constructive.
The two paths to $|1\rangle$ carry opposite arrows: destructive.

A qubit "flipped" twice

After one H : 50/50. After two: $|0\rangle$ with certainty. The second H does not re-randomise — it adds the arrows.

Same intermediate statistics, opposite final ones: the cross term is the entire difference.

Insert a phase between them: the outcome now reads an angle

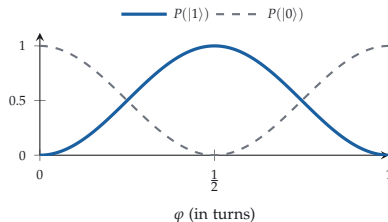


Now the lower path's arrow is turned by $2\pi\varphi$ before the paths meet:

$$A(|0\rangle) = \frac{1+e^{2\pi i \varphi}}{2}, \quad A(|1\rangle) = \frac{1-e^{2\pi i \varphi}}{2}.$$

Squaring, with $\Delta = 2\pi\varphi$ in Fact A:

$$P(|0\rangle) = \cos^2 \pi\varphi, \quad P(|1\rangle) = \sin^2 \pi\varphi.$$



A phase no measurement could see directly has become a probability we can sample.

Spread, write a phase, recombine: the whole algorithm, at one qubit. In Shor the phase will be s/r , and the second H becomes a Fourier transform on Q labels instead of 2 (Act VII).

Fact B — a unitary's eigenvalues are angles, its eigenvectors orthonormal

Eigenvalues are angles. If $U|u\rangle = \lambda|u\rangle$ with $\langle u|u\rangle = 1$, then

$$1 = \langle u|u\rangle = \langle Uu|Uu\rangle = |\lambda|^2 \implies \lambda = e^{2\pi i\varphi}, \varphi \in [0, 1).$$

φ is the **eigenphase**, measured in turns. A unitary can hand you no other kind of number.

Eigenvectors are orthogonal. If $\lambda \neq \lambda'$:

$$\langle u'|u\rangle = \langle Uu'|Uu\rangle = \overline{\lambda'}\lambda \langle u'|u\rangle,$$

and $\overline{\lambda'}\lambda \neq 1$ forces $\langle u'|u\rangle = 0$. In fact every unitary has a whole **orthonormal eigenbasis** $\{|u_k\rangle\}$, and in it U is diagonal:

$$U = \sum_k e^{2\pi i\varphi_k} |u_k\rangle\langle u_k|.$$

States are rays

$|u\rangle$ and $e^{i\theta}|u\rangle$ give the same probabilities for *every* measurement, since $|\langle e|e^{i\theta}u\rangle|^2 = |\langle e|u\rangle|^2$. They are the same physical state. So a physical state is a **ray**: a unit vector, up to an overall phase.

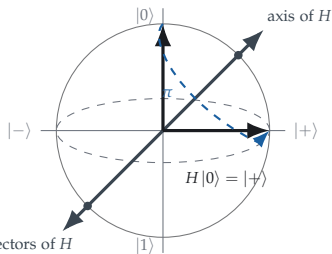
An eigenvector is an *eigen-ray*: U maps it to the same ray. The eigenbasis is a set of mutually orthogonal rays — states a measurement tells apart perfectly.

Both halves are used later

Angles: once the period is an eigenvalue, it *must* show up as a phase.

Orthogonality: branches tagged by different eigenvectors cannot interfere (Act VI).

One qubit, drawn: a unitary is a rotation, its eigenvectors are the axis



A qubit state, up to its overall phase (a ray, Fact B), is a point on a sphere:

$$\cos \frac{\theta}{2} |0\rangle + e^{i\gamma} \sin \frac{\theta}{2} |1\rangle \mapsto (\theta, \gamma) = (\text{latitude}, \text{longitude}),$$

and orthogonal states sit at *opposite* points.

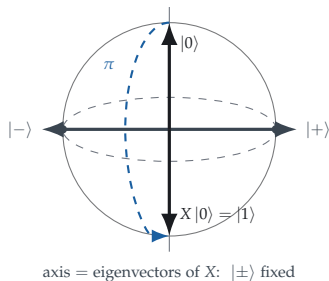
Every unitary is a rotation; its axis is its eigebasis

Every one-qubit unitary acts on the sphere as a rotation about some axis. A rotation moves every point except the two where the axis meets the sphere — and those two points are exactly **the eigenvectors of U** : U changes them only by a phase, their eigenvalues (Fact B). Opposite points: an orthonormal eigenbasis.

Drawn: H is the half-turn about the diagonal axis. It carries $|0\rangle$ to $|+\rangle$ and $|1\rangle$ to $|-\rangle$; only the two axis states stay put.

Unitaries are rotations, eigenvectors are the axis, and an angle is all a rotation can hand you. Next: the same picture for a permutation.

The smallest permutation, drawn: X , its axis, its diagonal form



In the computational basis X swaps the two labels — a permutation, entries 0 and 1, no phase anywhere. On the sphere it is the half-turn about the axis through $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$, which do not move:

$$X|+\rangle = |+\rangle, \quad X|-\rangle = -|-\rangle.$$

They are its eigenvectors. In the eigenbasis $\{|+\rangle, |-\rangle\}$ the *same* map is diagonal, and the change of basis is H :

$$X = \underbrace{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}_{\text{basis } |0\rangle, |1\rangle} = H \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}}_{\text{basis } |+\rangle, |-\rangle} H, \quad -1 = e^{2\pi i \cdot \frac{1}{2}}.$$

A permutation of labels in one basis is a set of phases in another. Act III does exactly this to multiplication by a : a permutation in the computational basis, a diagonal of angles $e^{2\pi i s/r}$ in its eigenbasis.

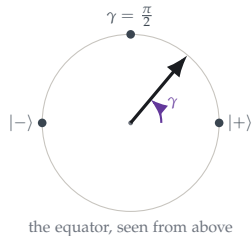
Fact C — why a relative phase is physical and a global phase is not

Global phase: every arrow turns together. $e^{i\theta} |\psi\rangle$ keeps every length and every angle *between* arrows: same ray, same point on the sphere, same statistics in every basis (Fact B).

Relative phase: one arrow turns against the others.

$$\frac{1}{\sqrt{2}} (|0\rangle + e^{i\gamma} |1\rangle)$$

gives $\frac{1}{2}, \frac{1}{2}$ in the computational basis for every γ . But each γ is a *different ray* — a different point on the equator — and another basis sees it: after H , $P(|0\rangle) = \cos^2 \frac{\gamma}{2}$.



Why it works

The relative phase is the *longitude*: it moves the state. A global phase moves nothing. A change of basis can turn the first into a probability, never the second.

Resolved. A global property can reach one observable event: interference moves probability before the sample is drawn (Fact A), and a relative phase is the handle a change of basis can grab (Fact C). The rest of the talk manufactures one such phase.

Act III — the operator: the period becomes an angle

Act III — multiplication by a is a permutation of the basis

Difficulty 3. No value of f contains r (Act I). Where does r live, in a form a physical process could couple to? Answer: in an *operator*.

First the register the operator acts on, then the operator itself.

The target register

$n = \lceil \log_2 N \rceil$ qubits: $2^n \geq N$ computational basis vectors

$$|w\rangle, \quad w = 0, 1, \dots, 2^n - 1,$$

one for every integer label — enough to give every residue mod N its own basis vector. ($N = 21$: $n = 5$, labels $0, \dots, 31$.)

The operator

Define U_a on *every* basis vector:

$$U_a |w\rangle = \begin{cases} |aw \bmod N\rangle, & 0 \leq w < N, \\ |w\rangle, & N \leq w < 2^n. \end{cases}$$

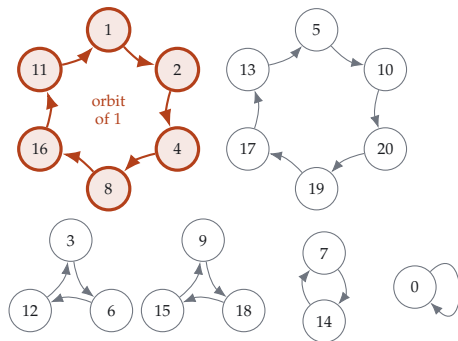
Here w is *any* label. Fixing U_a on each basis vector fixes it on every state (linearity, R2).

A permutation matrix

Since $\gcd(a, N) = 1$, $w \mapsto aw \bmod N$ is a bijection of $\{0, \dots, N-1\}$: U_a sends basis vectors to basis vectors, one-to-one. Its matrix has a single 1 in every row and column and 0s elsewhere — so it is unitary, and r appears in none of its entries.

Next: draw the permutation, and find r in its shape.

The permutation, drawn: $N = 21$, $a = 2$ splits into cycles



(labels 21, ..., 31: fixed, not drawn)

The 21 residues fall into **cycles** of lengths 6, 6, 3, 3, 2, 1.

- **Not every cycle has length r .** Labels sharing a factor with 21 (0; 7, 14; 3, 6, 12; 9, 18, 15) sit on shorter cycles.
- **Ours does.** We will always start from $|1\rangle$ (Act V). Its orbit $\mathcal{O} = \{1, a, a^2, \dots\}$ first returns to 1 at $a^r \equiv 1$: length exactly r , by the definition of the order. (The same holds from any w coprime to N : $a^k w \equiv w \iff a^k \equiv 1$.)
- **Restricted to the orbit, U_a is a cyclic shift:** $U_a|a^k\rangle = |a^{k+1}\rangle$, and $U_a^r = 1$ on $\mathcal{O}$.
- **Powers are permutations too:** $U_a^k = U_{a^k \bmod N}$ — on $\mathcal{O}$, a shift by k steps.

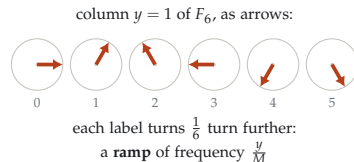
r is not an entry of the matrix: it is the length of the cycle through 1. The question “what is r ?” is now “how long is this loop?”

The tool: the Fourier matrix is a Vandermonde matrix of roots of unity

Pure linear algebra, any dimension M . Let $\omega = e^{2\pi i/M}$: the powers $1, \omega, \dots, \omega^{M-1}$ are the M points evenly spaced on the unit circle.

$$F_M = \frac{1}{\sqrt{M}} \begin{pmatrix} 1 & 1 & \dots & 1 \\ 1 & \omega & \dots & \omega^{M-1} \\ 1 & \omega^2 & \dots & \omega^{2(M-1)} \\ \vdots & \vdots & & \vdots \\ 1 & \omega^{M-1} & \dots & \omega^{(M-1)^2} \end{pmatrix}, \quad (F_M)_{xy} = \frac{\omega^{xy}}{\sqrt{M}}.$$

Vandermonde. Column y is the geometric sequence $(1, z, z^2, \dots, z^{M-1})/\sqrt{M}$ with $z = \omega^y$. A matrix whose columns are powers of chosen points is a *Vandermonde matrix*; here the points are the M -th roots of unity.



The columns are an orthonormal basis

$\langle \text{col}_{y'} | \text{col}_y \rangle = \frac{1}{M} \sum_x \omega^{x(y-y')} = D_M\left(\frac{y-y'}{M}\right)$: 1 if $y = y'$; otherwise the arrows walk once round a closed loop and the sum is 0 (Fact A). So F_M is unitary: the change of basis from labels to ramps.

A shift becomes a rotation

Let T be the cyclic shift of M labels, $T|x\rangle = |x+1 \bmod M\rangle$ — a permutation, like U_a on its orbit. Shift a geometric column $(1, z, z^2, \dots, z^{M-1})$ down one place:

$$T \begin{pmatrix} 1 \\ z \\ \vdots \\ z^{M-2} \\ z^{M-1} \end{pmatrix} = \begin{pmatrix} z^{M-1} \\ 1 \\ \vdots \\ z^{M-3} \\ z^{M-2} \end{pmatrix} = z^{-1} \begin{pmatrix} 1 \\ z \\ \vdots \\ z^{M-2} \\ z^{M-1} \end{pmatrix} \quad \text{because } z^M = 1 \text{ closes the loop.}$$

Every column of F_M is an eigenvector of T ; column y has eigenvalue $z^{-1} = \omega^{-y} = e^{-2\pi i y/M}$ — a turn by y/M of a full turn. In kets, the same computation reads

$$T \left(\frac{1}{\sqrt{M}} \sum_x \omega^{xy} |x\rangle \right) = \frac{1}{\sqrt{M}} \sum_x \omega^{xy} |x+1\rangle = \omega^{-y} \left(\frac{1}{\sqrt{M}} \sum_{x'} \omega^{x'y} |x'\rangle \right) \quad (x' = x+1),$$

$$T = F_M \text{diag}(1, \omega^{-1}, \omega^{-2}, \dots, \omega^{-(M-1)}) F_M^\dagger$$

In the label basis a shift moves labels. In the Fourier basis it moves nothing: it only turns each basis vector by its own angle. Shift $\rightarrow$ rotation.

Repeat the shift: powers become powers of the diagonal

Shift k times = turn k times as far

Write $T = FDF^\dagger$. The inner $F^\dagger F = \mathbf{1}$ cancel:

$$T^k = FDF^\dagger FDF^\dagger \cdots FDF^\dagger = F D^k F^\dagger.$$

A diagonal matrix is raised to a power entry by entry:

$$D^k = \text{diag}(\omega^{-yk})_y :$$

column y is turned by $k \cdot y/M$ turns. Every angle is a multiple of $1/M$, so $T^M = \mathbf{1}$: **the period of the shift is the denominator of its angles.**

Every circulant, at once

A **circulant** matrix is constant along its (wrapped) diagonals — a polynomial in T :

$$C = \begin{pmatrix} c_0 & c_2 & c_1 \\ c_1 & c_0 & c_2 \\ c_2 & c_1 & c_0 \end{pmatrix} = c_0 \mathbf{1} + c_1 T + c_2 T^2.$$

The same F diagonalises all of them:

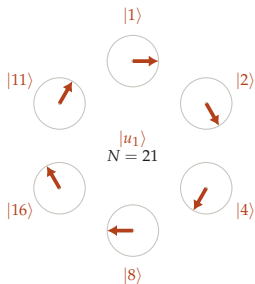
$$C = F \text{diag}(c(\omega^{-y})) F^\dagger, \quad c(z) = \sum_k c_k z^k.$$

Its eigenvalues are the polynomial c evaluated at the roots of unity — which is exactly what a Vandermonde matrix computes.

One matrix, three scales: $M = r$ diagonalises U_a on its orbit (next); $M = Q$ is the readout (Act VII); $M = 2$ is the Hadamard gate. Diagonalising a cyclic shift is the discrete Fourier transform.

On the orbit, label the basis vector $|a^k \bmod N\rangle$ by its exponent k : then U_a is the shift T with $M = r$. The previous slides give its eigenvectors at once (conjugated columns, so that the eigenvalue carries a $+$ sign):

$$|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i s k / r} |a^k \bmod N\rangle, \quad U_a |u_s\rangle = e^{2\pi i s / r} |u_s\rangle, \quad s = 0, \dots, r-1.$$



Check directly. $U_a |u_s\rangle = \frac{1}{\sqrt{r}} \sum_k e^{-2\pi i s k / r} |a^{k+1}\rangle$; rename $k+1 \rightarrow k$ (the loop closes since $a^r \equiv 1$) and a factor $e^{2\pi i s / r}$ comes out.

Picture. $|u_1\rangle$ at $N = 21$: one arrow per orbit label, each turned $\frac{1}{6}$ turn back from the previous — a ramp *around the cycle*.

No extra qubits. Each $|u_s\rangle$ is a combination of r computational basis vectors of the same n -qubit register: the target register already contains every eigenvector.

One operator, two bases, two matrices

A linear map does not come with a matrix; a matrix is the map *written in a chosen basis*. The same U_a on the orbit, $N = 21$, $a = 2$, $r = 6$, $\omega = e^{2\pi i/6}$:

	computational basis					
	$ 1\rangle$	$ 2\rangle$	$ 4\rangle$	$ 8\rangle$	$ 16\rangle$	$ 11\rangle$
$ 1\rangle$	0	0	0	0	0	1
$ 2\rangle$	1	0	0	0	0	0
$ 4\rangle$	0	1	0	0	0	0
$ 8\rangle$	0	0	1	0	0	0
$ 16\rangle$	0	0	0	1	0	0
$ 11\rangle$	0	0	0	0	1	0

A shift: entries 0 and 1. r is hidden in the *shape* — the size of the loop.

	eigenbasis					
	$ u_0\rangle$	$ u_1\rangle$	$ u_2\rangle$	$ u_3\rangle$	$ u_4\rangle$	$ u_5\rangle$
$ u_0\rangle$	1					
$ u_1\rangle$		ω				
$ u_2\rangle$			ω^2			
$ u_3\rangle$				ω^3		
$ u_4\rangle$					ω^4	
$ u_5\rangle$						ω^5

Diagonal: r written plainly in every entry, as angles s/r .

The change of basis between them is the Fourier matrix F_r . On the whole register U_a is block-diagonal, one block per cycle; only the block of the orbit of 1 will ever matter. And the eigenbasis is **orthonormal**, $\langle u_{s'} | u_s \rangle = \delta_{ss'}$ (Fact B): the $|u_s\rangle$ are perfectly distinguishable tags — which will stop different branches from interfering (Act VI), the same mechanism as decoherence (Act IX).

Resolved. “What is r ?” has become “what are this rotation’s angles?” — and nothing quantum has happened yet.

Act IV — making an angle measurable: kickback and the ladder

Difficulty 4. On its own eigenvector, $U_a^x |u_s\rangle = e^{2\pi i s x/r} |u_s\rangle$ multiplies the *whole* state by one number: a global phase, invisible in every basis for every x (Fact C). How do you measure a global phase?

Idea. Make it relative: apply U to only *one branch* of a superposition, and compare with a branch where it was not applied. That is what a **controlled** gate does.

The syntax of $c-U$, piece by piece

$$c-U = \underbrace{|0\rangle\langle 0|}_{\text{control is } |0\rangle} \otimes \underbrace{\mathbf{1}}_{\substack{\text{target:} \\ \text{do nothing}}} + \underbrace{|1\rangle\langle 1|}_{\substack{\text{control} \\ \text{is } |1\rangle}} \otimes \underbrace{U}_{\substack{\text{target:} \\ \text{apply } U}} = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & U \end{pmatrix} \text{ (blocks).}$$

- $|0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $|1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ are the **projectors** of Act II: each *selects* one branch of the control. They are not the numbers 0 and 1, nor the zero and identity matrices.
- $A \otimes B$ acts factor by factor: $(A \otimes B)(|c\rangle \otimes |w\rangle) = A|c\rangle \otimes B|w\rangle$ — left on the control, right on the target.
- So on basis vectors: $c-U|0\rangle|w\rangle = |0\rangle|w\rangle$, $c-U|1\rangle|w\rangle = |1\rangle U|w\rangle$. Example: CNOT = $c-X$.

Kickback: the phase moves to the control

Let $|u\rangle$ be an eigenvector, $U|u\rangle = e^{2\pi i\varphi}|u\rangle$, and put the control in any superposition $\alpha|0\rangle + \beta|1\rangle$. Each step uses one rule:

$$\begin{aligned} c\text{-}U\left[(\alpha|0\rangle + \beta|1\rangle) \otimes |u\rangle\right] &= \alpha|0\rangle \otimes |u\rangle + \beta|1\rangle \otimes U|u\rangle && \text{linearity, then the definition of } c\text{-}U \text{ on each branch} \\ &= \alpha|0\rangle \otimes |u\rangle + \beta|1\rangle \otimes (e^{2\pi i\varphi}|u\rangle) && |u\rangle \text{ is an eigenvector} \\ &= \alpha|0\rangle \otimes |u\rangle + (\beta e^{2\pi i\varphi}|1\rangle) \otimes |u\rangle && \text{bilinearity: the number moves to the other factor} \\ &= (\alpha|0\rangle + \beta e^{2\pi i\varphi}|1\rangle) \otimes |u\rangle && \text{bilinearity: factor out } |u\rangle \end{aligned}$$

Why bilinearity is the crux

The tensor product does not record which register a number “belongs to” (R4). The phase that U produced on the target can equally be read as a phase on the control’s $|1\rangle$ branch — and since the $|0\rangle$ branch received none, it is now a *relative* phase: visible (Fact C). The superposition of the control survives throughout.

Target	unchanged: $ u\rangle$ before and after. One copy serves every later controlled gate.
Control	gains the relative phase $e^{2\pi i\varphi}$: a turn of φ about the Bloch sphere’s vertical axis.
The pair	still a product state: nothing is entangled, because the target is an eigenvector.

Controlled powers for free: square-and-multiply, in superposition

Classically: square-and-multiply

To compute $a^x \bmod N$, write x in binary, $x = \sum_j x_j 2^j$:

$$a^x = \prod_j (c_j)^{x_j}, \quad c_j = a^{2^j} \bmod N.$$

The constants come from repeated squaring, $c_{j+1} = c_j^2 \bmod N$; then multiply by c_j exactly when bit $x_j = 1$ — the multiplicative form of double-and-add.

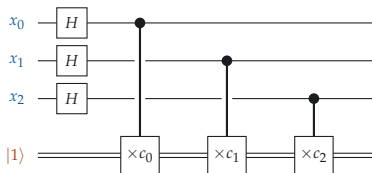
$N = 21, a = 2$: $c_0, c_1, c_2 = 2, 4, 16$. For $x = 5 = 101_2$:
 $2^5 \equiv c_0 c_2 = 32 \equiv 11$. ✓

Quantumly: the same program

Hold bit x_j in counting qubit j ; **rung** j multiplies the target by c_j , controlled by that qubit. Together the rungs form the **ladder** Λ :

$$\Lambda |x\rangle |w\rangle = |x\rangle |a^x w \bmod N\rangle,$$

and on a superposition of all x it runs every branch at once.

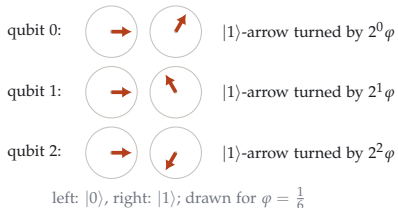


A control line passes *over* the wires it crosses; the dot marks the one qubit it reads.

The exponential trap, avoided: built by repetition, $c \cdot U_a^{2^{t-1}}$ would cost $2^{t-1} \approx N^2$ uses of U_a . But $U_a^{2^j} = U_{c_j}$ is one multiplication by a constant computed classically — t squarings on a laptop.

The ladder on an eigenvector: each qubit receives one digit of the angle

Feed the ladder an eigenvector, $U|u\rangle = e^{2\pi i\varphi}|u\rangle$. Rung j applies U^{2^j} , eigenvalue $e^{2\pi i 2^j \varphi}$: by kickback, qubit j turns by $2^j \varphi$ — twice as far as the qubit before. The target never moves.



After all rungs the counting register is a *product* of single qubits:

$$\bigotimes_j \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i 2^j \varphi} |1\rangle) = \frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} e^{2\pi i \varphi x} |x\rangle,$$

because branch x collects the turns of its 1-bits:
 $\prod_j e^{2\pi i 2^j \varphi x_j} = e^{2\pi i \varphi x}.$

The phase ramp

$|\Phi_\varphi\rangle := \frac{1}{\sqrt{Q}} \sum_x e^{2\pi i \varphi x} |x\rangle$ ($Q = 2^t$): each label's arrow turns φ further than the previous one. With $|\Phi_0\rangle = H^{\otimes t} |0 \cdots 0\rangle$, for Shor's eigenvectors: $\Lambda(|\Phi_0\rangle \otimes |u_s\rangle) = |\Phi_{s/r}\rangle \otimes |u_s\rangle.$

Resolved. The once-global phase now advances relatively, at rate s/r , across the counting register: a wave whose frequency is the answer — written in binary, one digit per rung.

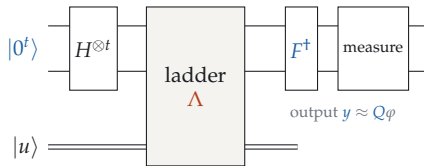
The phase-estimation problem — and the one ingredient Shor lacks

Phase estimation (Kitaev, 1995)

Given: (i) circuits for the controlled powers $c-U^{2^j}, j < t$; (ii) one copy of an eigenvector $|u\rangle$, $U|u\rangle = e^{2\pi i\varphi}|u\rangle$.

Find: φ , to about t binary digits.

Solution: $H^{\otimes t}$, then the ladder (this act), then a Fourier transform that reads the ramp's frequency (Act VII), then measure.



for Shor, $U = U_a$:

- | | |
|--------------------------------|--------------------------------|
| (i) controlled powers | ✓ cheap: $U_a^{2^j} = U_{c_j}$ |
| the phase φ | s/r : exactly what we want |
| (ii) eigenvector $ u_s\rangle$ | ? its formula contains r |
-

Recall $|u_s\rangle = r^{-1/2} \sum_k e^{-2\pi i s k / r} |a^k \bmod N\rangle$: to prepare the input, it seems we must already know the unknown.

Phase estimation solves “find the angle of a given eigenvector”. Shor must also solve “...without being able to write the eigenvector down”. That is Act V.

Act V — the input problem, and the $|1\rangle$ trick

Act V — the circularity, and the free input

Difficulty 5. Phase estimation needs an eigenvector $|u_s\rangle$ as input, and the formula for $|u_s\rangle$ contains r — *the very unknown*. Is the method circular?

No: the eigenvectors add up to a basis vector we can make with one gate:

$$|1\rangle = \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle$$

Proof. Insert the formula for $|u_s\rangle$ and swap the two finite sums:

$$\frac{1}{\sqrt{r}} \sum_s |u_s\rangle = \sum_k \left(\frac{1}{r} \sum_s e^{-2\pi i s k / r} \right) |a^k\rangle = |a^0\rangle = |1\rangle,$$

the inner sum being $D_r(-k/r)$: 1 at $k = 0$, else a closed walk, 0 (Fact A). $\square$

Smallest case: $N = 15$, $a = 4$, $r = 2$: $|u_{0,1}\rangle = \frac{1}{\sqrt{2}}(|1\rangle \pm |4\rangle)$, and $\frac{1}{\sqrt{2}}(|u_0\rangle + |u_1\rangle) = |1\rangle$.

No gate is told r : the decomposition of $|1\rangle$ is our analysis, true whether or not anyone computes it. Linearity makes the analysis binding.

What we get

$|1\rangle$ costs **one** X gate, and it contains all r eigenvectors with equal weight. By linearity (R2) the circuit then runs on every one of them at once. A spike in the orbit basis is flat in the eigenbasis.

And the price (Acts V–VI)

Each run hands back a *random* s , uniform on $\{0, \dots, r-1\}$, and the two registers end up entangled.

One rung on the real input: kickback becomes entanglement

$|1\rangle$ is *not* an eigenvector. Apply one rung, $N = 21$, $a = 2$, and write the *same* output vector in the two bases of Act III:

$$\begin{aligned} c-U_a \left[\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes |1\rangle \right] &= \frac{1}{\sqrt{2}} (|0\rangle |1\rangle + |1\rangle |2\rangle) && \text{computational basis: a } \textit{value} \text{ is written} \\ &= \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i s/r} |1\rangle) \otimes |u_s\rangle && \text{eigenbasis: } r \text{ kickbacks at once} \end{aligned}$$

Read it as a correlation

The target's value *depends* on the control's bit: $|1\rangle$ if the bit is 0, $|2\rangle$ if it is 1 — the correlation between x and $f(x)$ of Act I, for a one-bit x . A product state would need the *same* target vector in both branches; here the two are orthogonal, so measuring the target would reveal the control's bit. Not a product: **entangled** (R4).

Or as phases glued to tags

In the eigenbasis the same correlation reads: “control phase $s/r \leftrightarrow$ target tag $|u_s\rangle$ ”. Each branch alone is Act IV's kickback; the entanglement is that the phases differ from tag to tag. Averaged over the r tags, the phases cancel: on its own, the control shows no phase at all.

Same gate, two outcomes: on an eigenvector kickback entangles nothing (Act IV); on the free input it records the branch, and that record is the entanglement.

After the whole ladder: one state, three groupings

Run all t rungs on $|\Phi_0\rangle \otimes |1\rangle$. The result can be written in three exactly equal ways:

$$\underbrace{\frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} |x\rangle |f(x)\rangle}_{\substack{\text{by argument} \\ \text{the table of } f}} = \underbrace{\sum_{c=0}^{r-1} \sqrt{\frac{M_c}{Q}} |\psi_c\rangle \otimes |a^c \bmod N\rangle}_{\substack{\text{by value} \\ \text{Act I's combs}}} = \underbrace{\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |\Phi_{s/r}\rangle \otimes |u_s\rangle}_{\substack{\text{by frequency} \\ \text{Act IV's ramps}}}$$

Left = middle. Collect the x that give the same value $f(x) = a^c$: they form the comb $|\psi_c\rangle = M_c^{-1/2} \sum_j |c + jr\rangle$ with $M_c \approx Q/r$ teeth.

Middle = right. Expand $|1\rangle = r^{-1/2} \sum_s |u_s\rangle$ and apply Act IV's ramp equation to each eigenvector (linearity).

No gate wrote a phase

Every rung is a permutation matrix, entries 0 and 1. The phases in the right-hand form are what that permutation looks like in its own eigenbasis.

The registers are entangled: in the middle and right forms, r orthogonal tags each carry a different counting state.

Measuring the counting register now gives a uniformly random x — noise. The period sits in the combs' spacing, equivalently in the ramps' rates. Next slide: every amplitude, written out.

The whole evolution, written out: $N = 15$, $a = 7$, three counting qubits

$r = 4$ (powers 7, 4, 13, 1). A toy $t = 3$, $Q = 8$ — far below the real $Q > N^2$, but every amplitude fits on the slide. Rung constants $c_0, c_1, c_2 = 7, 4, 1$.

start $|0\rangle |1\rangle$

$$H^{\otimes 3} \frac{1}{\sqrt{8}} (|0\rangle + |1\rangle + |2\rangle + |3\rangle + |4\rangle + |5\rangle + |6\rangle + |7\rangle) |1\rangle$$

$$\Delta \frac{1}{\sqrt{8}} (|0\rangle |1\rangle + |1\rangle |7\rangle + |2\rangle |4\rangle + |3\rangle |13\rangle + |4\rangle |1\rangle + |5\rangle |7\rangle + |6\rangle |4\rangle + |7\rangle |13\rangle)$$

Regrouped by value: combs of spacing 4

$$\frac{1}{2} [|\psi_0\rangle |1\rangle + |\psi_1\rangle |7\rangle + |\psi_2\rangle |4\rangle + |\psi_3\rangle |13\rangle],$$

$$|\psi_c\rangle = \frac{1}{\sqrt{2}} (|c\rangle + |c+4\rangle).$$

Regrouped by frequency: ramps at rate $s/4$

$$\frac{1}{2} \sum_{s=0}^3 |\Phi_{s/4}\rangle \otimes |u_s\rangle, \quad |\Phi_{s/4}\rangle = \frac{1}{\sqrt{8}} \sum_{x=0}^7 i^{sx} |x\rangle,$$

$$|u_s\rangle = \frac{1}{2} (|1\rangle + (-i)^s |7\rangle + (-1)^s |4\rangle + i^s |13\rangle).$$

($c_2 = 7^4 \bmod 15 = 1$: rung 2 does nothing, as $r = 4$ divides 2^2 .) Every counting label still has probability $\frac{1}{8}$; Act VII finishes the example.

Resolved. The input we cannot write down is not needed: $|1\rangle$ contains all r eigenvectors, and linearity runs every branch at once. The price — entanglement, one branch per eigenvector — is the next act's subject.

Act VI — why one run samples exactly one eigenphase

Act VI — a two-branch warm-up: the tag decides

Difficulty 6. The free input fed *all r eigenvectors at once*; linearity produced all r ramps, each entangled with its tag. Why will the outcome law be a mixture over s — no cross terms between branches — rather than an interference pattern? What removes them: linearity? entanglement? orthogonality?

Start concrete: Act V's single rung, at $N = 21$, $a = 2$, left the registers in

$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle|1\rangle + |1\rangle|2\rangle), \quad \langle 1|2\rangle = 0.$$

Apply H to the counting qubit and measure it. Label y receives $\frac{1}{2}$ from branch 0 and $\pm\frac{1}{2}$ from branch 1 — amplitudes attached to *different tags*, so what sits at an outcome is **not a number but a vector**, and (R3) takes its squared *norm*:

$$|\chi_0\rangle = \frac{1}{2}(|1\rangle + |2\rangle), \quad |\chi_1\rangle = \frac{1}{2}(|1\rangle - |2\rangle) \quad \implies \quad P(y) = \|\chi_y\|^2 = \frac{1}{4} + \frac{1}{4} = \boxed{\frac{1}{2} \text{ each.}}$$

Remove the tag, and interference returns

Had the work register stayed $|1\rangle$ in *both* branches: $|\chi_0\rangle = |1\rangle$, $|\chi_1\rangle = 0$, so $P = (1, 0)$ — Act II's H - H example. **The tag alone turned $(1, 0)$ into $(\frac{1}{2}, \frac{1}{2})$** , without touching the counting qubit.

The general two-branch law

For any V and any tags, $P(y) = |a_y|^2 + |b_y|^2 + 2 \operatorname{Re}(\overline{a_y} b_y \langle u_0 | u_1 \rangle)$. Orthogonal tags multiply the cross term by 0: the branches were added as *vectors*, but their *probabilities* add.

Orthogonal work-register tags $\Rightarrow$ no cross terms $\Rightarrow$ probabilities add. Shor is exactly this computation, with r orthogonal tags instead of two.

Why one run samples one eigenphase

Act V 's state, with its r orthogonal tags, is the warm-up at scale: $|\Psi\rangle = r^{-1/2} \sum_s |\Phi_{s/r}\rangle \otimes |u_s\rangle$. Apply *any* unitary V to the counting register (Act VII will choose $V = F^\dagger$) and measure it. The vector sitting at outcome y is $|\chi_y\rangle = r^{-1/2} \sum_s \langle y|V|\Phi_{s/r}\rangle |u_s\rangle$, and orthonormality of the tags removes every $s \neq s'$ term from its norm:

$$P(y) = \|\chi_y\|^2 = \sum_{s=0}^{r-1} \frac{1}{r} \left| \langle y|V|\Phi_{s/r}\rangle \right|^2 \quad \text{for every choice of } V.$$

These are exactly the statistics of: **draw s uniformly at random, then apply V to the single ramp $|\Phi_{s/r}\rangle$ and measure.** Eigenphases are sampled. Concretely, at $N = 21$ ($r = 6$, $Q = 1024$) with $V = F^\dagger$: six lobes, one per branch — the histogram of Act VIII — and the outcome $y = 171$ draws essentially all of its probability from the $s = 1$ branch.

Superposition globally, mixture locally

Nothing “randomly chose an eigenvector”: the joint state remains the one superposition $|\Psi\rangle$ of all r branches. But every statistic of the counting register alone is a uniform average over branches, because the branches carry orthogonal tags: **the counting register by itself behaves like a mixture of the r ramps.**

The one run you perform samples one eigenphase. (Quantum information packages this as the reduced density matrix of C ; that notation is in the backup slides and is not needed today.)

Where the entanglement lives — a register is a place, not a value

Every claim of the form “ C is in ...” needs a time-stamp:

after	joint state	C alone, W alone	pair
initialisation	$ 0\rangle^{\otimes t} 1\rangle$	each has its own ket	product
$H^{\otimes t}$	$ \Phi_0\rangle 1\rangle$	each has its own ket	product
the ladder Δ	$r^{-1/2} \sum_s \Phi_{s/r}\rangle u_s\rangle$	neither has a ket of its own	entangled , rank r
$F^\dagger$	$(F^\dagger \otimes \mathbf{1}) \Psi\rangle$	neither	entangled
measure C	$ y\rangle \tilde{w}_y\rangle / \ \tilde{w}_y\ $	each has its own ket again	product again

Rows 1–2 vs. rows 3–4

In rows 3–4 the joint state does not factor as $|a\rangle \otimes |b\rangle$ (R4): only the *pair* has a state vector. That is the content of entanglement, and it happens at the ladder and only there.

$F^\dagger$ touches only C

So nothing it does can change any W -statistic: no signalling, in miniature. And measuring W in the eigenbasis first — or never — leaves every C -statistic unchanged: both give the previous slide’s mixture formula.

Resolved. *Orthogonality of the tags (Fact B) deletes every cross term between branches: one run behaves as one s drawn uniformly. A which-branch record kills interference by existing, read or not — entanglement is not a nuisance here; it is the sampling mechanism.*

Act VII — the readout: the angle becomes a position

Difficulty 7. After the ladder every counting amplitude has the same modulus, $|\langle x | \Phi_\varphi \rangle|^2 = 1/Q$: measuring now returns noise. The answer sits in the *phases*; the Born rule reads only *moduli*. Needed: the unitary converting *which frequency* into *which label* — exactly, off-grid included.

Act III's matrix at $M = Q$, on the counting register: $F|y\rangle = |\Phi_{y/Q}\rangle$. The columns of F are the ramps, so $F^\dagger$ sends each grid ramp back to one label; for an arbitrary frequency φ , the amplitude on y is Fact A's geometric sum:

$$\langle y | F^\dagger | \Phi_\varphi \rangle = \frac{1}{Q} \sum_{x=0}^{Q-1} e^{2\pi i(\varphi - y/Q)x} = D_Q\left(\varphi - \frac{y}{Q}\right).$$

Read it as a bank of matched filters

Each label y is the hypothesis “the frequency is y/Q ”, and $F^\dagger$'s factor $e^{-2\pi ixy/Q}$ unwinds the ramp at that trial rate. Right hypothesis: Q unit phasors add coherently. Wrong: the residual phases sweep the circle and cancel.

$F^\dagger$ creates no probability

It *moves* it: from Q equal heaps into r lobes. Each conditional $|D_Q|^2$ sums to 1 over y by Parseval — constructive interference on the truth, destructive everywhere else.

A grid-aligned ramp ($\varphi = y/Q$) maps to one single basis label. An off-grid ramp spreads over nearby labels with the profile $|D_Q|^2$ — deterministically, and for a reason that has nothing to do with quantum mechanics.

The linearity of QPE: eigenphases are sampled

The whole quantum part is **one fixed linear map plus one measurement**, $\text{QPE} = (F^\dagger \otimes \mathbf{1}) \Delta (H^{\otimes t} \otimes \mathbf{1})$ — and nothing in it is told r . On a *single* eigenvector it is solved exactly: $\text{QPE}(|0^t\rangle \otimes |u_s\rangle) = \sum_y D_Q(\frac{s}{r} - \frac{y}{Q}) |y\rangle \otimes |u_s\rangle$ (Act IV's ramp, then the kernel above). The real input is not an eigenvector — but it is a *known combination* of them, $|1\rangle = r^{-1/2} \sum_s |u_s\rangle$, so

$$\text{QPE}(|0^t\rangle \otimes |1\rangle) \stackrel{\text{linearity}}{=} \frac{1}{\sqrt{r}} \sum_s \text{QPE}(|0^t\rangle \otimes |u_s\rangle) = \frac{1}{\sqrt{r}} \sum_s \sum_y D_Q(\frac{s}{r} - \frac{y}{Q}) |y\rangle |u_s\rangle,$$

and measuring C returns y with the squared *norm* of its component (R3) — Act VI's computation, at $V = F^\dagger$ — where orthogonality of the tags deletes every $s \neq s'$ term:

$$P(y) = \frac{1}{r} \sum_{s,s'} \overline{D_Q(\frac{s'}{r} - \frac{y}{Q})} D_Q(\frac{s}{r} - \frac{y}{Q}) \underbrace{\langle u_{s'} | u_s \rangle}_{\delta_{ss'}} = \boxed{\sum_{s=0}^{r-1} \underbrace{\frac{1}{r}}_{P(s)} \underbrace{|D_Q(\frac{s}{r} - \frac{y}{Q})|^2}_{P(y|s)}}$$

The marked equality is the only quantum mechanics used; expanding the input over eigenvectors is bookkeeping, and **linearity makes the bookkeeping binding**. Linearity delivers all r branches; the Born rule returns one label per run; orthogonality (Fact B) removes the cross terms — so the statistics are *exactly*: draw s uniformly, then run ideal phase estimation on s/r . **Eigenphases are sampled, never averaged.**

Were the tags linearly independent but not orthogonal, one run would still return one label — but $P(y)$ would carry interference between different s , and the sampling reading that Act VIII relies on would be false.

The filter bank, drawn: a matched filter per label, a peak per harmonic s/r

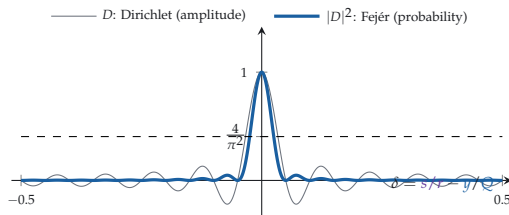
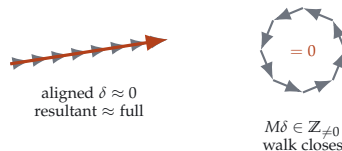
$D_M(\delta) = \frac{1}{M} \sum_x e^{2\pi i \delta x}$ is a walk in which every step turns by the same angle $2\pi\delta$.

Aligned ($\delta \rightarrow 0$): the hands point the same way and the resultant is nearly full length — *constructive*.

Misaligned ($M\delta \in \mathbb{Z}$ but $\delta \notin \mathbb{Z}$): the walk closes into whole loops and the resultant is *exactly zero* — *destructive*.

That is the entire mechanism: the walk is the response of the filter tuned to y/Q when fed the ramp of frequency φ , $\delta = \varphi - y/Q$. The amplitude is Dirichlet (sign-changing); the *probability* is its square, the non-negative Fejér profile.

The period enters as its r harmonics s/r : each answers the filters tuned nearest Qs/r , and *which filters fire* is what recovers r .



Two regimes, and why the leakage is not a quantum effect

Exact: $r \mid Q$

Every ramp $|\Phi_{s/r}\rangle$ is exactly a column of F , so F^+ returns a single label with certainty:

$$P(y = \frac{Qs}{r}) = \frac{1}{r}, \quad P(y) = 0 \text{ otherwise.}$$

Verified at $N = 15, a = 7$: $r = 4 \mid Q = 256$; the outcome is exactly $1/4$ at $y = 0, 64, 128, 192$ and exactly 0 elsewhere. *This is the textbook toy — and it is a toy, because $r \mid Q$ almost never happens.*

Real: $r \nmid Q$

The ramp is off-grid; D_Q spreads it. The integer y nearest Qs/r (so $|\delta| \leq 1/2Q$) still gets

$$P(y \mid s) = |D_Q(\delta)|^2 \geq \frac{4}{\pi^2} \approx 40.5\%.$$

Jordan's inequality: $|\sin \pi Q \delta| \geq 2Q|\delta|$ and $|\sin \pi \delta| \leq \pi|\delta|$; the bound is attained in the limit at $\delta = 1/2Q$.

“Only 40%? Add qubits.”

No. Qubits buy *resolution* (the main lobe narrows as $1/Q$), not *sidelobes*: $Q \int |D_Q|^2 = 1$ always, while $Q \int |D_Q| \sim \frac{4}{\pi^2} \ln Q \rightarrow \infty$. The tails are what a finite window *is*: to raise the floor, taper the window — or just repeat.

Resolved. *The angle became a position: y lands nearest Qs/r with probability at least $4/\pi^2$ per branch, and exactly on Qs/r with certainty when $r \mid Q$. What leaks is the transform of a finite rectangular window — a Fourier fact, not an error — and repetition is cheaper than tapering.*

The $N = 15$ example, finished: $F^\dagger$ and one measurement

Act V left $\frac{1}{2} \sum_c |\psi_c\rangle \otimes |7^c\rangle$ with combs $|\psi_c\rangle = \frac{1}{\sqrt{2}}(|c\rangle + |c+4\rangle)$. Here $r = 4$ divides $Q = 8$: the exact regime. $F^\dagger$ turns each comb of spacing $r = 4$ into a comb of spacing $Q/r = 2$, and its offset c into phases:

$$F^\dagger |\psi_c\rangle = \frac{1}{2} \sum_{\ell=0}^3 (-i)^{c\ell} |2\ell\rangle \quad (\text{the odd labels cancel exactly}).$$

Collect by counting label — each line is one outcome:

$$\begin{aligned} & |0\rangle \otimes \frac{1}{4} (|1\rangle + |7\rangle + |4\rangle + |13\rangle) \\ & + |2\rangle \otimes \frac{1}{4} (|1\rangle - i|7\rangle - |4\rangle + i|13\rangle) \\ & + |4\rangle \otimes \frac{1}{4} (|1\rangle - |7\rangle + |4\rangle - |13\rangle) \\ & + |6\rangle \otimes \frac{1}{4} (|1\rangle + i|7\rangle - |4\rangle - i|13\rangle) \end{aligned} = \frac{1}{2} \sum_{\ell=0}^3 |2\ell\rangle \otimes |u_\ell\rangle.$$

Each target vector has squared length $\frac{1}{4}$: the outcomes $y = 0, 2, 4, 6$ each have probability $\frac{1}{4}$, and labels 1, 3, 5, 7 never occur. The target is left in $|u_\ell\rangle$ — by value *and* by frequency, the same answer.

Reading the outcome

$y/Q = \ell/4$:

$y = 2$	$\frac{1}{4}$	$7^4 \equiv 1$: $r = 4$ ✓
$y = 6$	$\frac{3}{4}$	denominator 4 ✓
$y = 4$	$\frac{1}{2}$	$7^2 \equiv 4 \neq 1$: reject
$y = 0$	0	no information

Success in half the runs; failures are detected by the classical check. At the real size $Q = 256$ the same four outcomes appear, at $y = 0, 64, 128, 192$.

And the factors (Act X)

$7^2 \equiv 4$: $\gcd(4-1, 15) = 3$, $\gcd(4+1, 15) = 5$.

Act VIII — from one sample to the period: continued fractions

Act VIII — recovering an exact fraction from one noisy sample

Difficulty 8. The measurement returns an integer y with $y/Q \approx s/r$ — a fraction with a huge denominator approximating an unknown fraction with a small one, where *both* s and r are unknown. Recover the exact s/r from one sample — and how large must Q be for the answer to be unambiguous?

A nearest-bin outcome y satisfies $|\frac{y}{Q} - \frac{s}{r}| \leq \frac{1}{2Q} < \frac{1}{2N^2} < \frac{1}{2r^2}$, so one theorem applies:

Legendre's theorem

If $\gcd(s, r) = 1$ and $|\xi - s/r| < 1/2r^2$, then s/r is a *convergent* of the continued fraction of ξ .

So: expand y/Q by the Euclidean algorithm (polynomial time), take the convergent denominators $q < N$, and **verify** $a^q \equiv 1 \pmod{N}$ **classically** — a wrong answer is never mistaken for a right one. (The test alone proves only $r \mid q$; on the good event $q \mid r$ too, so passing forces $q = r$.)

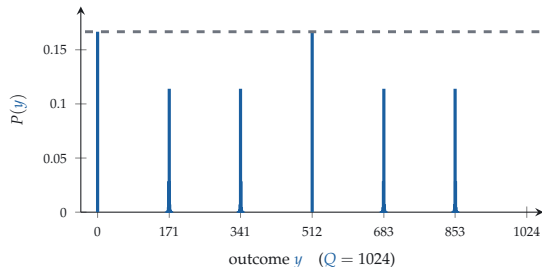
Two failure modes, both detected

$s = 0$ (probability $1/r$) returns $y = 0$: no information. $\gcd(s, r) > 1$ returns a proper divisor of r . The fraction of s coprime to r is $\Omega(1/\log \log r)$: budget a handful of runs, or take the lcm of two.

Two independent arguments give the same Q

Legendre needs $Q > r^2$ with r possibly near N ; and distinct candidates $s/r \neq s'/r'$ are $\geq 1/N^2$ apart, so resolution $1/Q$ must beat $1/N^2$. Both give $t = 2n$: *the largest register in the machine is sized by the classical post-processing.*

The worked example: $N = 21$, $a = 2$ — and here $r \nmid Q$



Six lobes at $y \approx Qs/r = 0, 171, 341, 512, 683, 853$ — the $r = 6$ branches made visible.

Dashed line: $1/r$.

$n = 5$, $t = 10$, $Q = 1024$, $r = \text{ord}_{21}(2) = 6$, and $6 \nmid 1024$.

Take $s = 1$: $Qs/r = 170.\bar{6}$, nearest bin $y = 171$,
 $\delta = -1/3072$, so $P(171 \mid s=1) = 0.6839$ and
 $P(170 \mid s=1) = 0.1710$.

Post-processing. Expand $171/1024 = [0; 5, 1, 84, 2]$; the
 convergents are $0/1, 1/5, 1/6, 85/509$. Denominator
 5 fails ($2^5 = 32 \equiv 11$); **denominator 6 works**
 ($2^6 = 64 \equiv 1$): $r = 6$.

Resolved. One sample, Euclid's expansion, one classical check: $r = 6$. A run with $s = 2$ lands at $y = 341$ and gives denominator $3 = r/2$; the check $2^3 \not\equiv 1$ rejects it — redraw, or take $\text{lcm}(3, 6) = 6$. Wrong outcomes are never mistaken for right ones.

Act IX — scratch records: $P(y)$ depends on every register, measured or not

Act IX — the master formula, and its two extremes

Difficulty 9. So far exactly two records of the counting label exist: the label itself, and the value $|a^x \bmod N\rangle$. A real reversible multiplier also leaves *carry chains, comparison flags, partial products* — scratch that depends on x . What does an unread extra register do to $P(y)$? It had better be nothing — is it?

Let the pre-transform state be $|\Psi\rangle = \sum_x \alpha_x |x\rangle \otimes |g_x\rangle$, where $|g_x\rangle$ collects *everything else* — unit vectors, not necessarily orthogonal. Apply any unitary V on C alone and measure (R3):

$$P(y) = \left\| \sum_x V_{yx} \alpha_x |g_x\rangle \right\|^2 = \sum_{x,x'} V_{yx} \overline{V_{yx'}} \alpha_x \overline{\alpha_{x'}} \langle g_{x'} | g_x \rangle$$

Every unmeasured register enters $P(y)$: the diagonal terms $x = x'$ are the classical part, and each interference term $x \neq x'$ is multiplied by the overlap $\langle g_{x'} | g_x \rangle$ of that register's records.

Uncomputed: $|g_x\rangle \equiv |g\rangle$

The overlap is 1 with the subscript *absent*, so it factors out and the sum moves **inside** the modulus: $P(y) = |\sum_x V_{yx} \alpha_x|^2$ — full interference, the exact law of Act VII.

Perfect record: $\langle g_{x'} | g_x \rangle = \delta_{xx'}$

Every cross term dies, the sum moves **outside** the modulus, and since every entry of $V = F^\dagger$ has $|V_{yx}|^2 = 1/Q$, $P(y) = \sum_x |V_{yx}|^2 |\alpha_x|^2 = 1/Q$: **exactly flat**.

The failure mode of a dirty ancilla is not a wrong answer but a flat histogram. In between, damage is linear in the overlap — and a record leaking one bit within each coset halves the peak, exactly.

Two records, opposite rules — and Bennett's undo

In Shor the pre-transform state is really $\sum_x \alpha_x |x\rangle |a^x \bmod N\rangle |g_x\rangle_{\text{scr}}$, so the master formula carries *two* overlap factors:

$$P(y) \propto \sum_{x,x'} e^{2\pi i(x-x')y/Q} \underbrace{\langle a^{x'} | a^x \rangle}_{= [x \equiv x' \bmod r]} \underbrace{\langle g_{x'} | g_x \rangle}_{\text{must be } \equiv 1}$$

The work register's record is *wanted*

$[x \equiv x' \bmod r]$ separates only branches that must *not* interfere (different eigenbranches) and is blind within each coset. **The blind spot is the period: the signal is the blindness pattern of a which-path record.** *W* is never uncomputed, never measured.

Scratch is the tax

Carries and flags vary *within* a coset: they separate branches that must add coherently, killing exactly the cross terms that carry *r*. Both uniform policies fail: “unread is harmless” gives the flat $1/Q$; “uncompute everything” (*W* too) gives $|\Phi_0\rangle \otimes |0\rangle$, and $F^\dagger |\Phi_0\rangle = |0\rangle$: $y = 0$ with certainty.

Bennett's escape — reset is not unitary, so run the writing steps backwards: compute, *copy the answer out* with CNOTs (a basis-state copy is not cloning), uncompute. Every scratch wire returns to $|0\rangle$ in every branch, the overlaps back to 1. **Half of Shor's arithmetic exists to undo the other half.**

Resolved. *A register may survive to the transform iff its contents are a function of the coset label alone. Decoherence is the same equation, with the environment as the register you failed to uncompute.*

Act X — the finale: why the period factors N

Act X — the classical reduction, saved for last

Difficulty 10. The machine now delivers $r = \text{ord}_N(a)$ for any coprime base a . One classical step remains: why was the *period* worth having? Turn r into a factor of N .

The reduction, in one line

If r is even and $a^{r/2} \not\equiv -1 \pmod{N}$, then $x = a^{r/2}$ satisfies $x^2 \equiv 1$ with $x \not\equiv \pm 1$ ($x \not\equiv 1$ by *minimality* of r), so $N \mid (x-1)(x+1)$ while N divides neither factor, and

$$\gcd(a^{r/2} \pm 1, N) \text{ are non-trivial factors of } N.$$

For N odd, composite, not a prime power, both conditions hold with probability $\geq \frac{1}{2}$ over random a — and $\geq 1 - 2^{-(m-1)}$ for m distinct prime factors.

$N = 21$, **finished**: $r = 6$ even, $2^3 = 8 \not\equiv -1 \pmod{21}$ ($\equiv 20$); $\gcd(7, 21) = 7$, $\gcd(9, 21) = 3$: $\boxed{21 = 3 \times 7}$. $N = 15$, $a = 7$: $r = 4$, $7^2 \equiv 4 \not\equiv \pm 1$; $\gcd(3, 15) = 3$, $\gcd(5, 15) = 5$: $\boxed{15 = 3 \times 5}$.



Resolved. The hardest case, $N = pq$, is exactly the case cryptography uses: $m = 2$ gives the familiar “at least half”. A failed draw of a is detected and redrawn.

The complete algorithm

1. **Even? Perfect power?** If $2 \mid N$, return 2; if $N = b^c$, $c \geq 2$, return b (test $c \leq \log_2 N$ integer roots — necessary: the reduction fails on prime powers).
2. **Lucky?** Draw $a \in \{2, \dots, N-1\}$ uniformly; if $\gcd(a, N) > 1$, return it — a factor for free.
3. **Precompute** $c_j = a^{2^j} \bmod N$ for $j = 0, \dots, t-1$ by repeated squaring ($t = 2n$; classical, $O(n^3)$).
4. **Quantum order finding:** prepare $|0\rangle^{\otimes t} |1\rangle$ (one X); apply $H^{\otimes t}$; apply the ladder of controlled $\times c_j$; apply $F^\dagger$ to C ; measure $C \rightarrow y$.
5. **Recover:** continued fractions on y/Q ; candidate denominator q ; verify $a^q \equiv 1 \pmod{N}$; else repeat, or lcm-combine two runs.
6. **Usable?** If r odd or $a^{r/2} \equiv -1$: back to step 2. Else **return** $\gcd(a^{r/2} \pm 1, N)$.

The pipeline in one line: $f(x+r) = f(x) \Rightarrow U_a^r = \mathbf{1}$ on the orbit $\Rightarrow \varphi_s = s/r \xrightarrow{c \cdot U_a^{2^j}} e^{2\pi i x s / r} \xrightarrow{F^\dagger} y \approx \frac{Q}{r} s \xrightarrow{\text{CF}} r \xrightarrow{\gcd} p, q$.

The whole run — one vector, two readings at every step

after	computational reading (values are written)	eigenbasis reading (phases turn)	C : W
init	$ 0^t\rangle 1\rangle$	$\frac{1}{\sqrt{r}} \sum_s 0^t\rangle u_s\rangle$	product
$H^{\otimes t}$	$\frac{1}{\sqrt{Q}} \sum_x x\rangle 1\rangle = \Phi_0\rangle 1\rangle$	$\frac{1}{\sqrt{r}} \sum_s \Phi_0\rangle u_s\rangle$	product
the ladder Λ	$\frac{1}{\sqrt{Q}} \sum_x x\rangle f(x)\rangle$ (period = repetition)	$\frac{1}{\sqrt{r}} \sum_s \Phi_{s/r}\rangle u_s\rangle$ (period = r rates)	entangled, rank r
$F^+ \otimes 1$	$\sum_y y\rangle \otimes \tilde{w}_y\rangle, \quad \tilde{w}_y\rangle = \frac{1}{Q} \sum_x e^{-2\pi i xy/Q} f(x)\rangle$	$\frac{1}{\sqrt{r}} \sum_s \sum_y D_Q(\frac{s}{r} - \frac{y}{Q}) y\rangle u_s\rangle$	entangled
measure $C \rightarrow y$	$ y\rangle \otimes \tilde{w}_y\rangle / \ \tilde{w}_y\ , \quad P(y) = \ \tilde{w}_y\ ^2$	$ y\rangle \otimes \sum_s c_s u_s\rangle, \quad c_s ^2 = P(s \mid y)$	product again

The left column is a machine that *writes values*: the period enters as literal repetition, $f(x + r) = f(x)$, no phase appears anywhere below F^+ , and interference survives exactly within the cosets. The right column is a machine that *turns phases*: W never moves, the period enters as the r rates s/r , and F^+ reads each rate as a label. **Each row is one vector written twice** — the same $Q \cdot 2^n$ complex numbers — which is why Kitaev’s phase-estimation network came out identical to Shor’s period-finding network.

The measured W -weights are the Bayesian posterior: collapse is conditioning. At $Q = 1024$, $r = 6$, the outcome $y = 171$ pins the branch, $P(s=1 \mid y=171) = 0.999993$.

The same run in closed form ($r \mid Q$): three groupings, end to end

Exact case, so every $M_c = Q/r$. (The general case and the by-argument grouping after $F^\dagger$, $\sum_y |y\rangle \otimes |\tilde{w}_y\rangle$, are on the previous slide.)

$$\begin{aligned}
 |0^t\rangle \otimes |1\rangle &\xrightarrow{H^{\otimes t}} |\Phi_0\rangle \otimes |1\rangle \xrightarrow{\Lambda} \underbrace{\frac{1}{\sqrt{Q}} \sum_x |x\rangle |f(x)\rangle}_{\text{by argument}} = \underbrace{\frac{1}{\sqrt{r}} \sum_{c=0}^{r-1} |\psi_c\rangle \otimes |a^c\rangle}_{\text{by value (cosets)}} = \underbrace{\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |\Phi_{s/r}\rangle \otimes |u_s\rangle}_{\text{by frequency (ramps)}} \\
 &\xrightarrow{F^\dagger \otimes 1} \underbrace{\frac{1}{r} \sum_{c,\ell} e^{-2\pi i \ell c / r} |\ell Q / r\rangle \otimes |a^c\rangle}_{\text{by value: the offset } c \text{ survives only as a unimodular phase}} = \underbrace{\frac{1}{\sqrt{r}} \sum_s |s Q / r\rangle \otimes |u_s\rangle}_{\text{by frequency: each ramp returns to its label}}
 \end{aligned}$$

The step on the left is $F^\dagger |\psi_c\rangle = r^{-1/2} \sum_\ell e^{-2\pi i \ell c / r} |\ell Q / r\rangle$ — Act VII's kernel with every off-grid label cancelling exactly. Group the left sum over c : since $r^{-1/2} \sum_c e^{-2\pi i \ell c / r} |a^c\rangle = |u_\ell\rangle$, the two sides agree term by term — the identification $\ell = s$ is the definition of the eigenvector. Measuring C gives $P(y = \ell Q / r) = \frac{1}{r}$ on both sides (left: $\sum_c r^{-2}$; right: $|r^{-1/2}|^2$), and leaves W in $|u_\ell\rangle$.

Shor's sampled label ℓ and phase estimation's sampled eigenphase ℓ/r are the same integer, met from opposite directions. When $r \nmid Q$, the deltas $|\ell Q / r\rangle$ widen into the Dirichlet lobes of Act VII; nothing else changes.

The three things to keep

1. The answer is spectral, and the input is free

r is not in any value of f , and not in any entry of U_a . It is the length of a cycle, hence an eigenphase. And $|1\rangle = r^{-1/2} \sum_s |u_s\rangle$ — the state you cannot prepare is not needed, because the state you get for **one X gate** contains all r of them.

2. Branches interfere only if they agree on *every* register

Measured or not. One leftover scratch bit costs half the peak; a fully distinguishable record of *which input* x flattens the histogram to $1/Q$; a stray photon is the same equation; and a record of *which eigenbranch* s is the opposite case — the sampling mechanism. This is why half of Shor's arithmetic undoes the other half, and why fault tolerance is not optional.

3. The arithmetic, not the Fourier transform, is the machine

For RSA-2048: $\approx 2.7 \times 10^9$ Toffolis of reversible modular multiplication against $\approx 10^5$ gates of approximate QFT — the transform is about 0.004% of the circuit.

In one line: the period becomes an angle, the angle becomes a position, and continued fractions turn the position back into the period.

Shor's algorithm, in a single paragraph

The period r of $f(x) = a^x \bmod N$ is a global property: every value of f is cheap, and no list of sampled values reveals r . So make it the property of an operator: $U_a|w\rangle = |aw \bmod N\rangle$ is a permutation whose entries are all 0s and 1s, yet on the orbit of $|1\rangle$ it is a cyclic shift of length exactly r , and — being unitary — the only numbers it can offer are the angles $e^{2\pi i s/r}$: *the period now sits in a spectrum*. On its own eigenvector U_a produces a global phase, which no measurement in any basis can see; applied *controlled* on a counting register in uniform superposition, kickback makes the angle a *relative* phase, and doubling the controlled power rung by rung — affordable because $U_a^{2^j} = U_{a^{2^j \bmod N}}$, one modular constant per rung — writes it across the register as a wave $|\Phi_{s/r}\rangle$ *whose frequency is the answer*.

The eigenvector itself cannot be prepared, since its formula contains r , and it need not be: $|1\rangle = r^{-1/2} \sum_s |u_s\rangle$ costs one X gate, and linearity runs the circuit on all r eigenbranches at once. The price is entanglement: the counting register alone then behaves as a *mixture* of the r waves, because the tags $|u_s\rangle$ are orthogonal and no interference survives between branches — each run behaves as a single s drawn uniformly, never a blur. Measuring at this point would return the flat distribution, since every amplitude of a wave has modulus $Q^{-1/2}$ and the Born rule reads only moduli; so the one remaining gate is the inverse Fourier transform, whose columns *are* these waves: it turns the frequency s/r into the labels nearest Qs/r , the nearest with probability at least $4/\pi^2$ (the remainder is the leakage of a finite window — a Fourier effect, not an error). Continued fractions applied to a good outcome's y/Q recover s/r in lowest terms; the classical check $a^q \equiv 1 \pmod{N}$ accepts the recovered denominator exactly when it is the period, and a handful of runs makes success near-certain. The period was worth the trouble because of one classical line: for N odd, composite and not a prime power, if r is even and $a^{r/2} \not\equiv -1 \pmod{N}$ — probability at least $\frac{1}{2}$ over random a — then N divides $(a^{r/2} - 1)(a^{r/2} + 1)$ while dividing neither factor, so $\gcd(a^{r/2} \pm 1, N)$ are nontrivial factors. **Nowhere was the period computed: the circuit changed basis until the period could be read off the label a measurement returns — and Euclid finished the job.**

Backup — every symbol, one meaning

N, a	modulus; base, $\gcd(a, N) = 1$	$ u_s\rangle$	eigenvector of U_a , eigenvalue $e^{2\pi i s/r}$
r	the order $\text{ord}_N(a)$ — <i>the unknown</i>	s	eigenbranch index, $s = 0, \dots, r-1$
$f(x)$	$a^x \bmod N$	$ \Phi_\varphi\rangle$	phase ramp $Q^{-1/2} \sum_x e^{2\pi i \varphi x} x\rangle$
n	target qubits, $n = \lceil \log_2 N \rceil$	$ \psi_c\rangle$	comb $M_c^{-1/2} \sum_j c + jr\rangle$
t, Q	counting qubits $t = 2n$; $Q = 2^t > N^2$	c_j	rung constant $a^{2^j} \bmod N$
C, x, y	counting register; input / measured labels	Λ	the ladder of controlled multiplications
W, w	target register; its labels	$F, F^\dagger$	Fourier transform on C , and inverse
U_a	$ w\rangle \mapsto aw \bmod N\rangle$	$D_M(\delta)$	Dirichlet kernel $\frac{1}{M} \sum_{x < M} e^{2\pi i \delta x}$
$\mathcal{O}$	orbit $\{1, a, \dots, a^{r-1}\}$ of 1	$c\text{-}U$	$ 0\rangle\langle 0 \otimes \mathbf{1} + 1\rangle\langle 1 \otimes U$

Backup — density matrices: the packaging we skipped

For a pure state, $\rho = |\psi\rangle\langle\psi|$; for a pair, the **reduced state** of C is $\rho_C = \text{tr}_W |\Psi\rangle\langle\Psi|$, computed by $\text{tr}_W(A \otimes B) = A \text{ tr } B$: the one operator that returns every C -statistic, $P(y) = \langle y | V \rho_C V^\dagger | y \rangle$.

The smallest example: a Bell pair

$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ has $\rho_A = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{1}{2}\mathbf{1}$: the pair has a perfectly definite state vector, yet either qubit alone is a fair coin. A subsystem of a pure state can fail to have a ket of its own — exactly the situation after Shor's ladder.

For the post-ladder state, expanding tr_W with $\langle u_{s'} | u_s \rangle = \delta_{ss'}$:

$$\boxed{\rho_C = \frac{1}{r} \sum_{s=0}^{r-1} |\Phi_{s/r}\rangle\langle\Phi_{s/r}|} \iff (\rho_C)_{xx'} = \frac{[x \equiv x' \bmod r]}{Q} \quad \underset{Q=6, r=3}{=} \frac{1}{6} \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$

The diagonal is flat at $1/Q$ — measuring before $F^\dagger$ returns noise — and the period sits entirely in *which off-diagonal coherences survive*: those with $x \equiv x' \pmod{r}$.

Stated precisely: for a pure state of a pair, the reduced state of either register is mixed ($\text{tr } \rho^2 < 1$) exactly when the pair is entangled. And discarding a register, or measuring it and forgetting the outcome, leave the same reduced state — the other register's statistics cannot distinguish the two.

Backup — gates, and where each sits in Shor

Below F^+ there are only two kinds of gate: **permutations of labels** (entries 0/1, no phase — the arithmetic) and **diagonal phase writers** (populations untouched). H is the only gate here that *converts* between them, because it is the Fourier transform on $\mathbb{Z}_2$.

gate	computational action	what it does	where in Shor
X	$ 0\rangle \leftrightarrow 1\rangle$	swaps labels	prepares $ 1\rangle$ (once)
Z	$\text{diag}(1, -1)$	writes a relative phase	the ladder's phases are Z-like
H	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	delta $\leftrightarrow$ flat; $H = F_2$	makes the Q branches
CNOT	$ x\rangle 0\rangle \mapsto x\rangle x\rangle$	copies a <i>basis</i> state; correlates	copy-out in Bennett; the leak that kills the fringes
Toffoli	reversible AND	the only nonlinear gate needed	every carry in the adder
c-U	$ 0\rangle\langle 0 \otimes \mathbf{1} + 1\rangle\langle 1 \otimes U$	on an eigenvector: rotates the <i>control</i> only	every rung of the ladder
U_a	$2^n \times 2^n$ permutation	cyclic shift of the orbit; r invisible	the operator being estimated
F	$(F)_{xy} = Q^{-1/2} \omega_Q^{xy}$ with $\omega_Q = e^{2\pi i/Q}$	sends a frequency <i>label</i> y to the ramp $ \Phi_{y/Q}\rangle$	so F^+ reads a ramp's frequency back as a label

Everything in Shor is: permute (arithmetic), phase (kickback), convert (H / F^+).

Backup — what it costs to build

The arithmetic tower

- L1 CDKM ripple-carry adder: $2n$ Toffolis, depth $O(n)$, one ancilla.
- L2 Modular add of a constant: a quantum computer cannot branch, so the overflow if becomes arithmetic — five constant additions in seven blocks, of which **blocks 5–7 exist only to un-write the flag**.
- L3 Out-of-place $\times c$: shift-and-add.
- L4 Controlled in-place $\times c$: multiply into scratch, rename registers, subtract the multiple back out (possible because $a^{-1} \bmod N$ exists).
- L5 Shor: $2n$ rungs, $\approx 4n + 2$ qubits.

$F^\dagger$ is nearly free

The product formula factorises it into one Hadamard per qubit plus $\binom{t}{2}$ controlled rotations $R_m = \text{diag}(1, e^{2\pi i/2^m})$. Rotations with $m \gtrsim \log t$ are ≈ 1 and can be dropped (Coppersmith): $O(t \log t)$ gates. Or skip the transform entirely — measure one qubit at a time with classically-controlled rotations (semiclassical QFT), recycling a single counting qubit.

RSA-2048 vs. the elliptic curve

Gidney–Ekerå: 2048-bit RSA in $\approx 2.7 \times 10^9$ Toffolis, $\approx 6,200$ logical qubits (~ 8 h, 20M physical). Litinski: a 256-bit elliptic-curve private key in $\approx 5 \times 10^7$ Toffolis — roughly $50\times$ cheaper.

The elliptic-curve systems are the *cheaper* target for a quantum attacker, not the safer one.